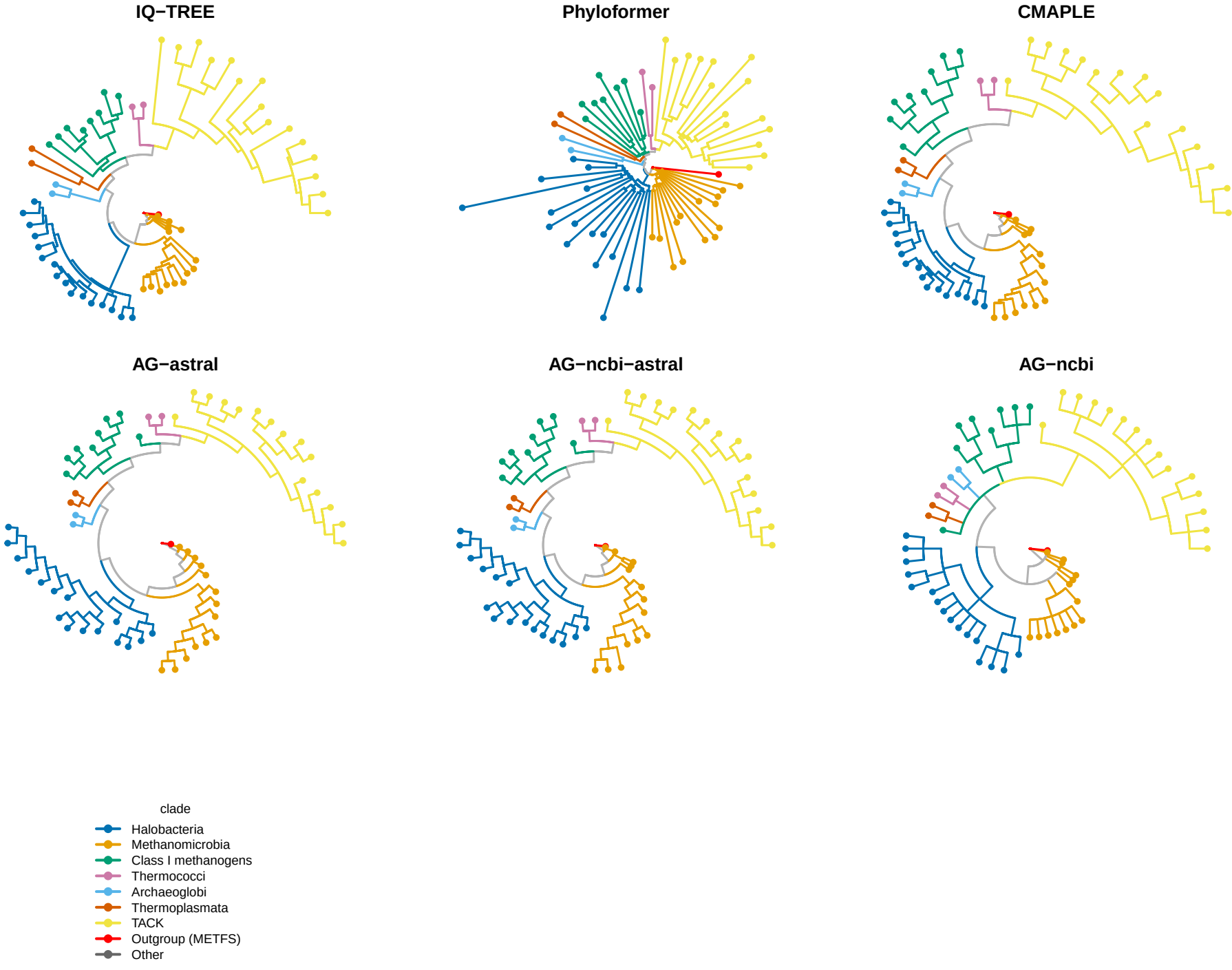
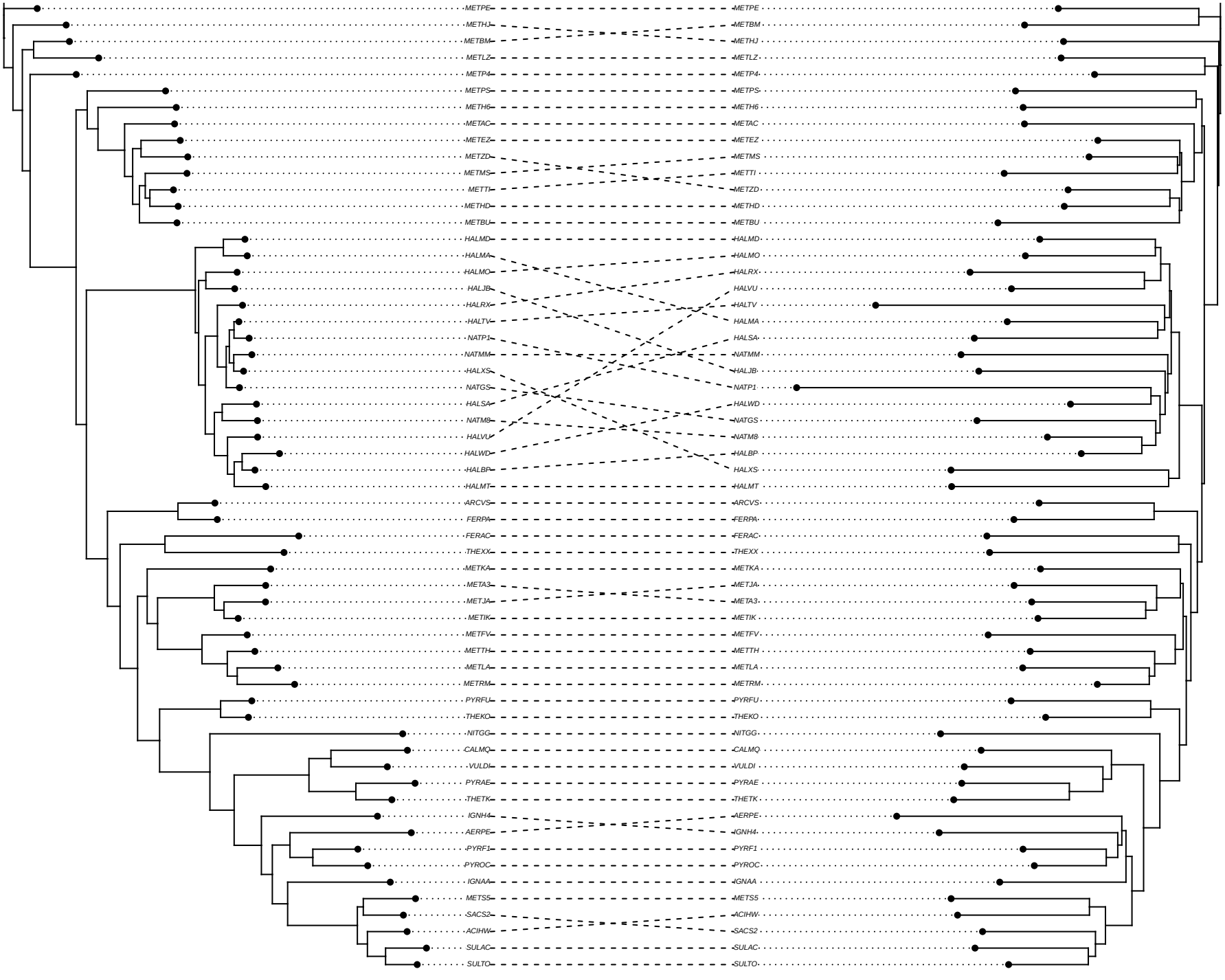
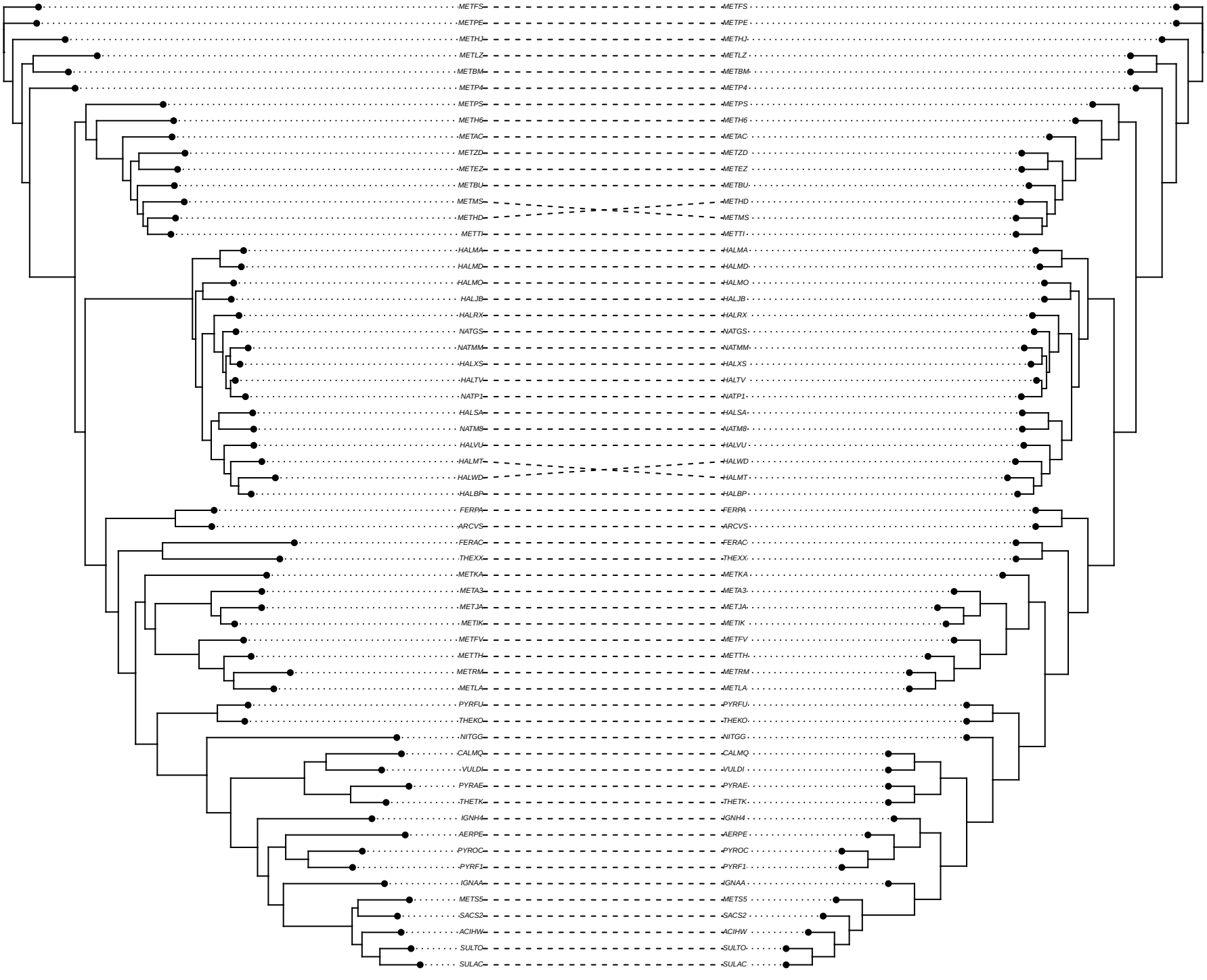
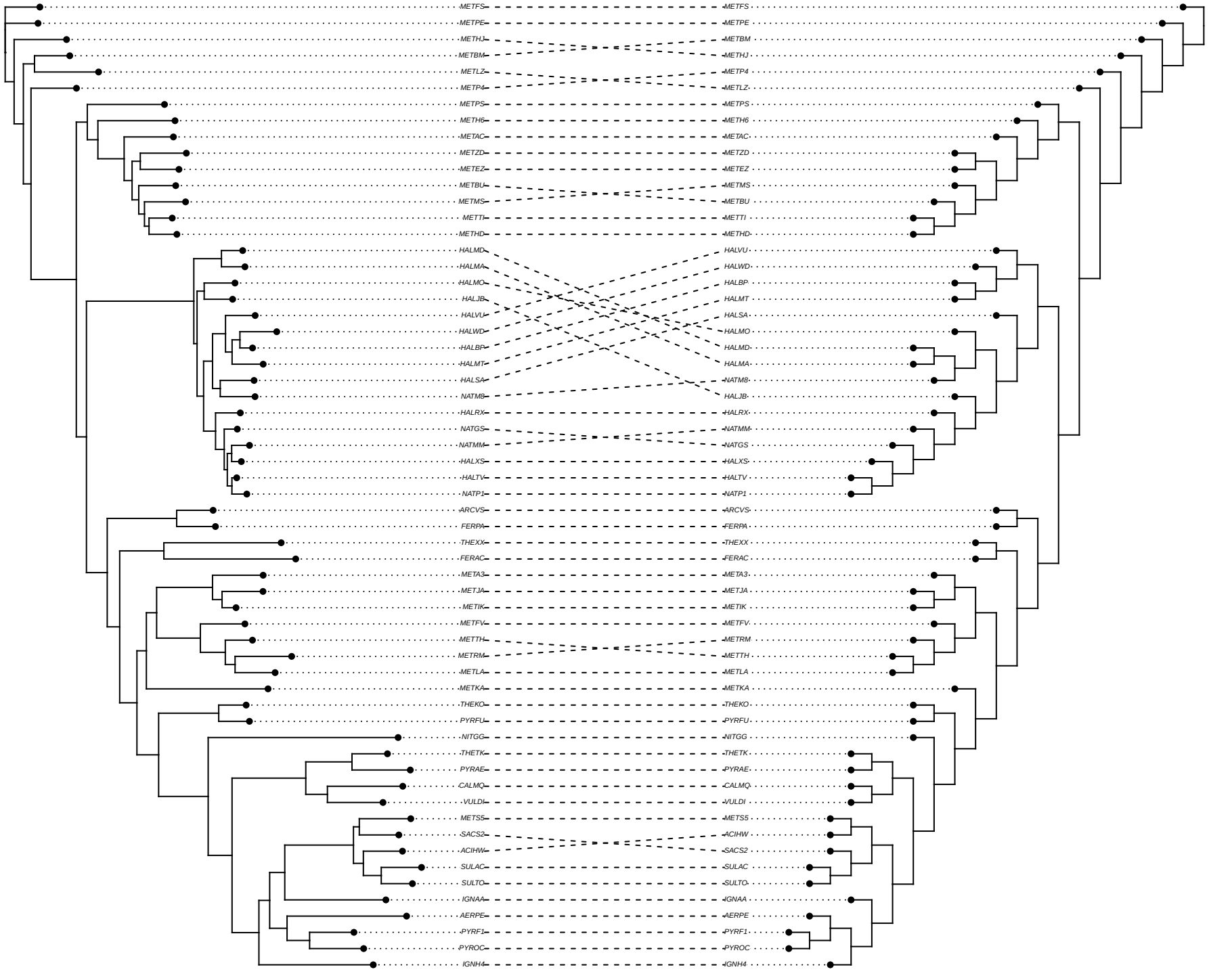

IQ−TREE
Phyloformer
CMAPLE
AG−astral
AG−ncbi−astral
AG−ncbi
clade
Halobacteria
Methanomicrobia
Class I methanogens
Thermococci
Archaeoglobi
Thermoplasmata
TACK
Outgroup (METFS)
Other


IQ−TREE vs Phylolformer
METPE
METHJ
METBM
METLZ
METP4
METPS
METH6
METAC
METEZ
METZD
METMS
METTI
METHD
METBU
HALMD
HALMA
HALMO
HALJB
HALRX
HALTV
NATP1
NATMM
HALXS
NATGS
HALSA
NATM8
HALVU
HALWD
HALBP
HALMT
ARCVS
FERPA
FERAC
THEXX
METKA
META3
METJA
METIK
METFV
METTH
METLA
METRM
PYRFU
THEKO
NITGG
CALMQ
VULDI
PYRAE
THETK
IGNH4
AERPE
PYRF1
PYROC
IGNAA
METS5
SACS2
ACIHW
SULAC
SULTO
METPE
METBM
METHJ
METLZ
METP4
METPS
METH6
METAC
METEZ
METMS
METTI
METZD
METHD
METBU
HALMD
HALMO
HALRX
HALVU
HALTV
HALMA
HALSA
NATMM
HALJB
NATP1
HALWD
NATGS
NATM8
HALBP
HALXS
HALMT
ARCVS
FERPA
FERAC
THEXX
METKA
METJA
META3
METIK
METFV
METTH
METLA
METRM
PYRFU
THEKO
NITGG
CALMQ
VULDI
PYRAE
THETK
AERPE
IGNH4
PYRF1
PYROC
IGNAA
METS5
ACIHW
SACS2
SULAC
SULTO

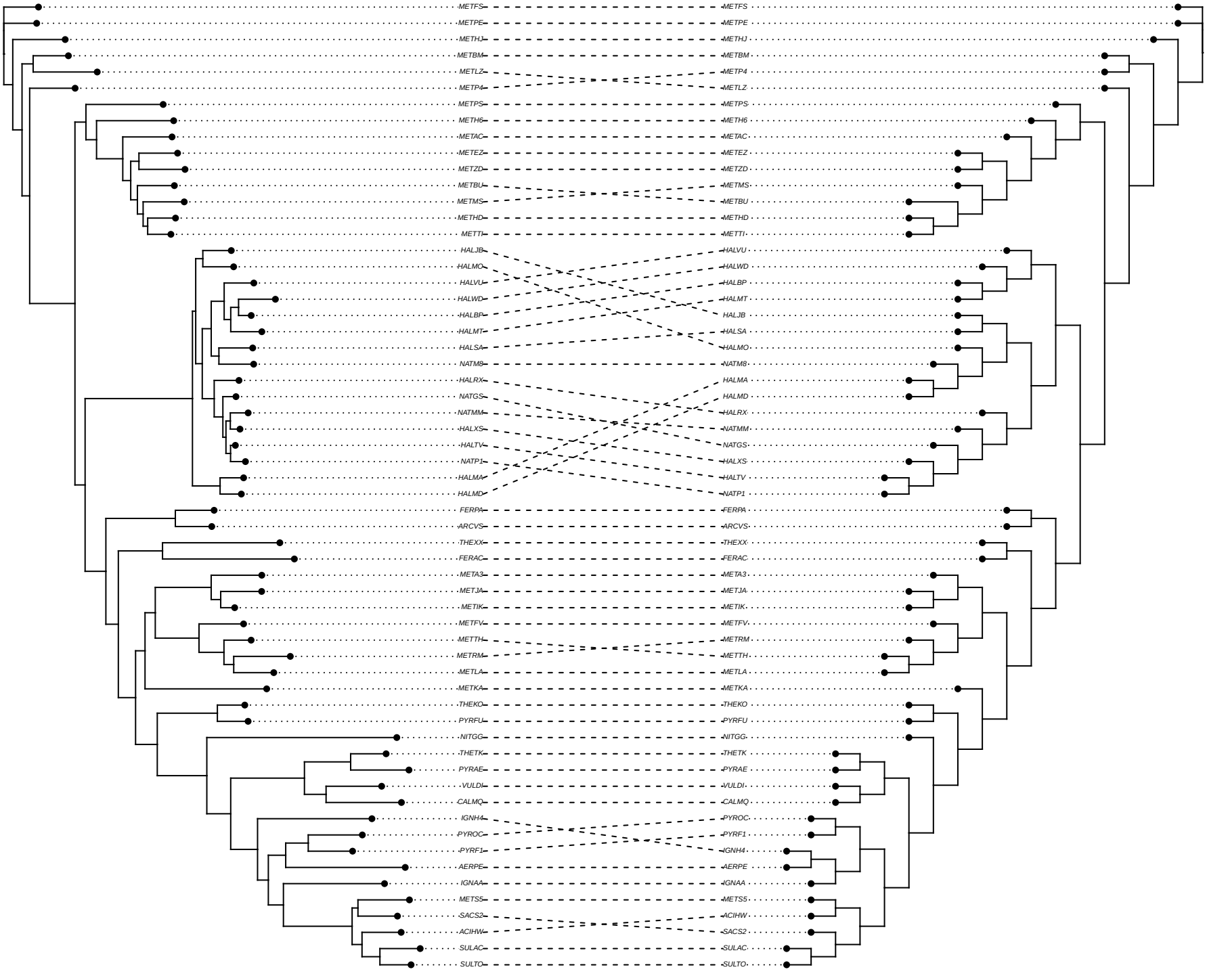
IQ-TREE vs CMAPLE



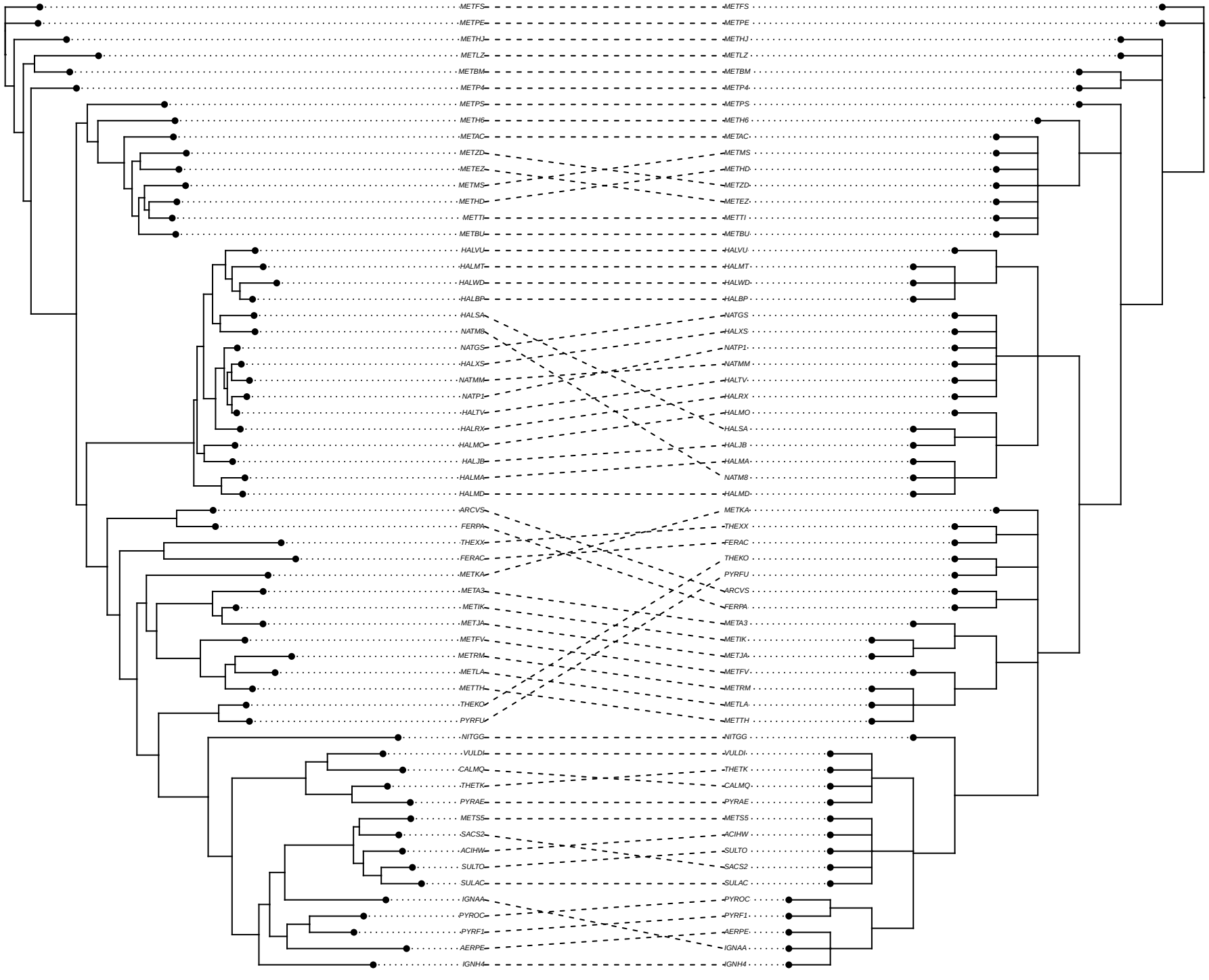
IQ−TREE vs AG−astral

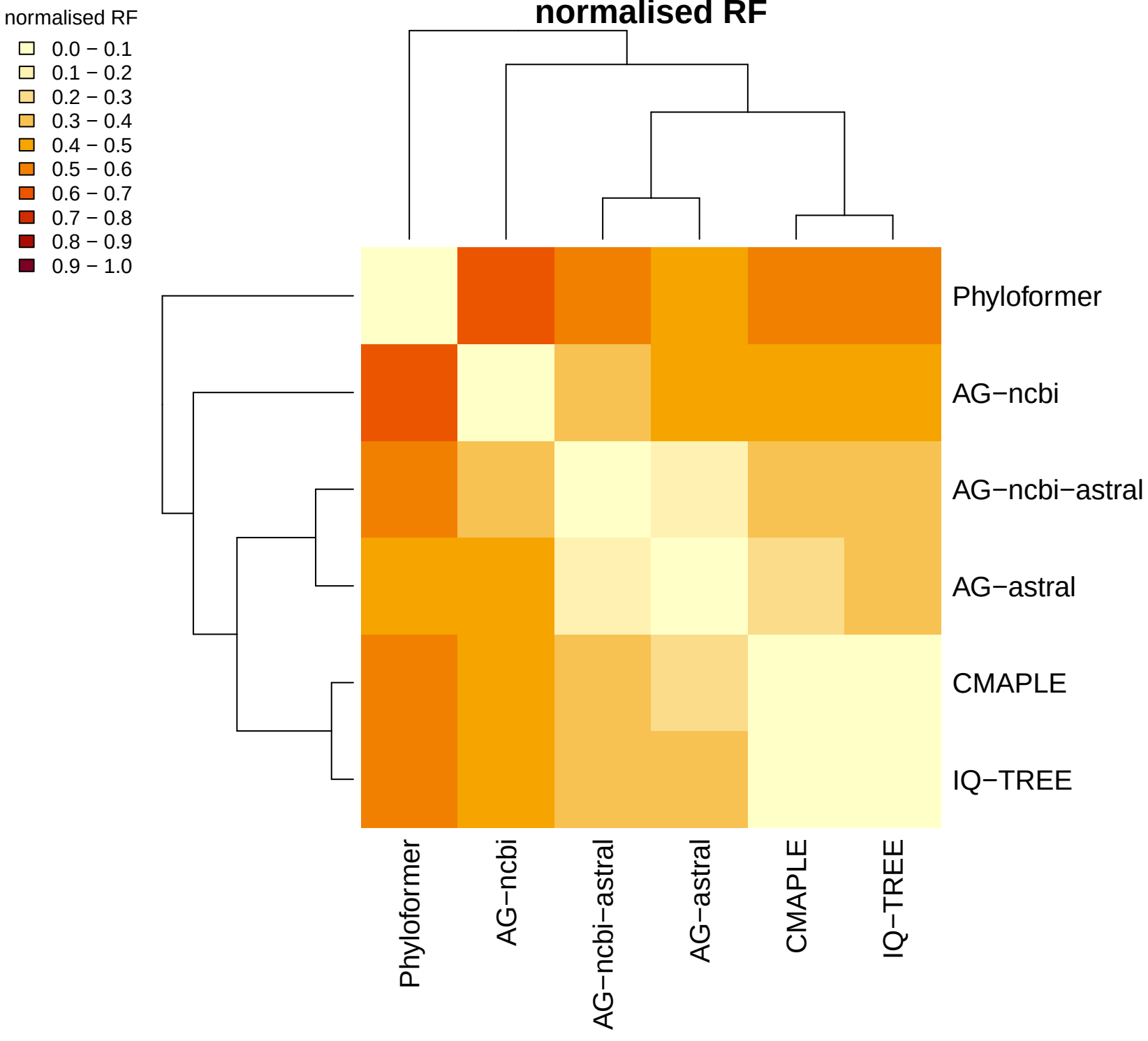


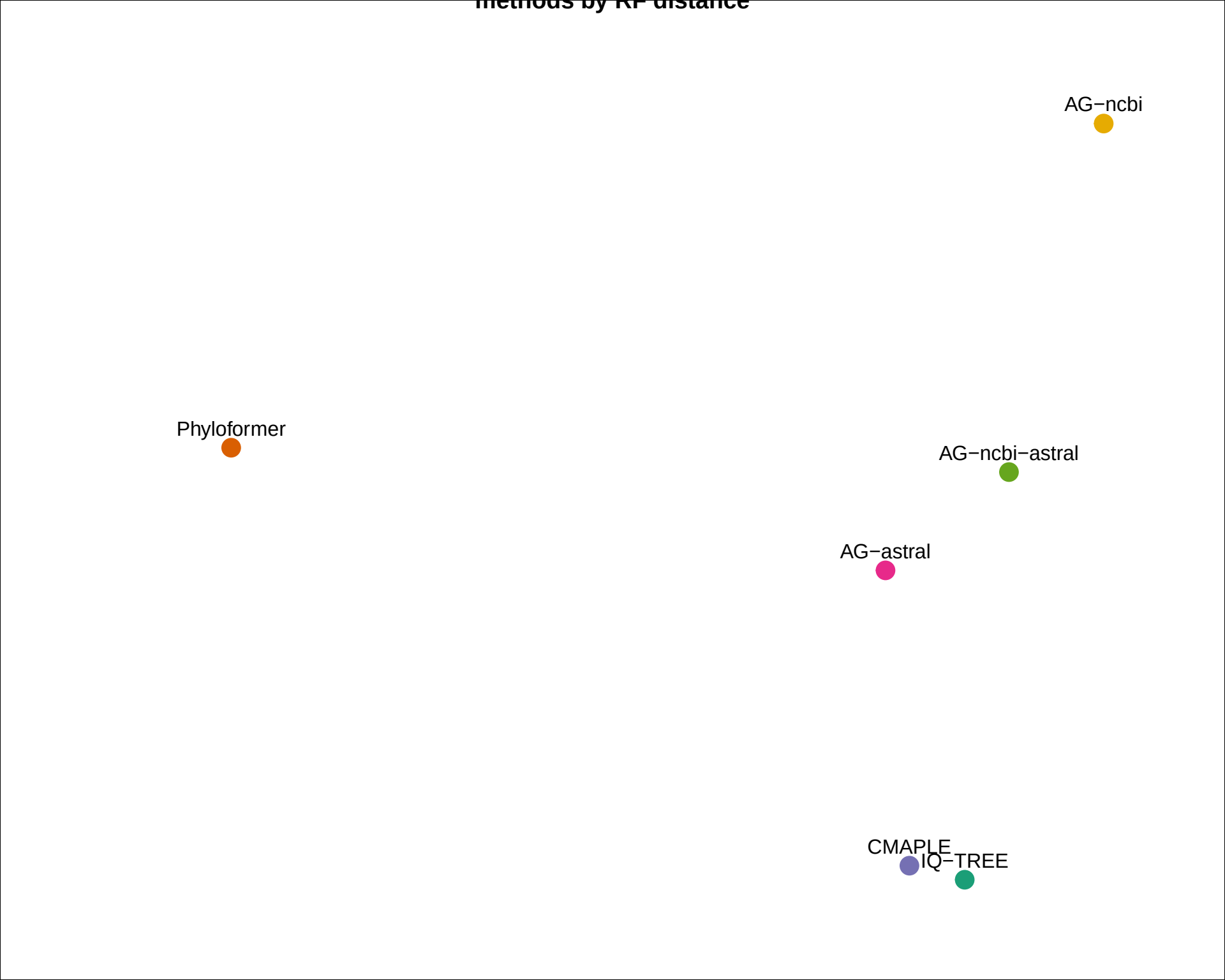
IQ-TREE vs AG-ncbi-astral

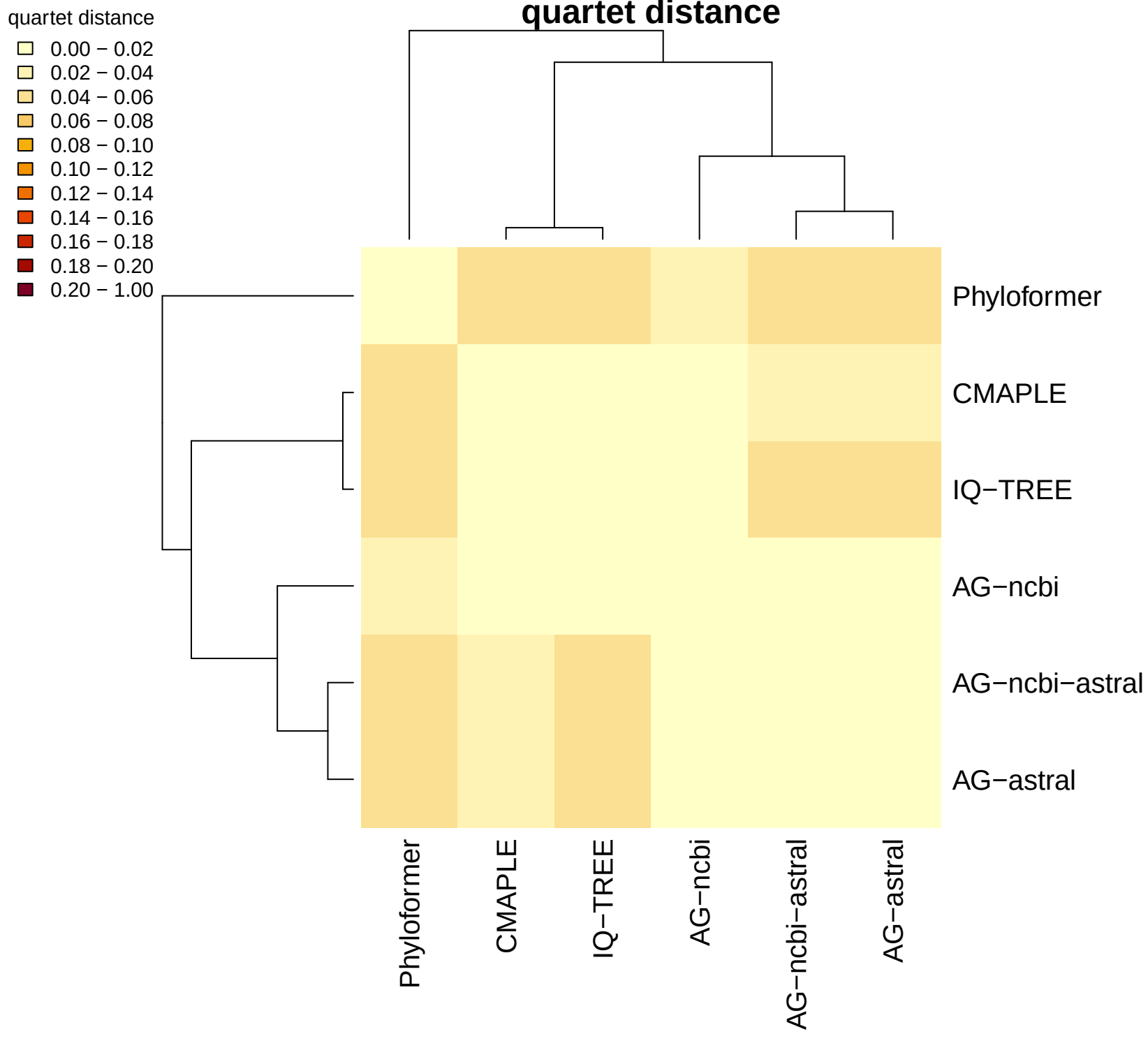


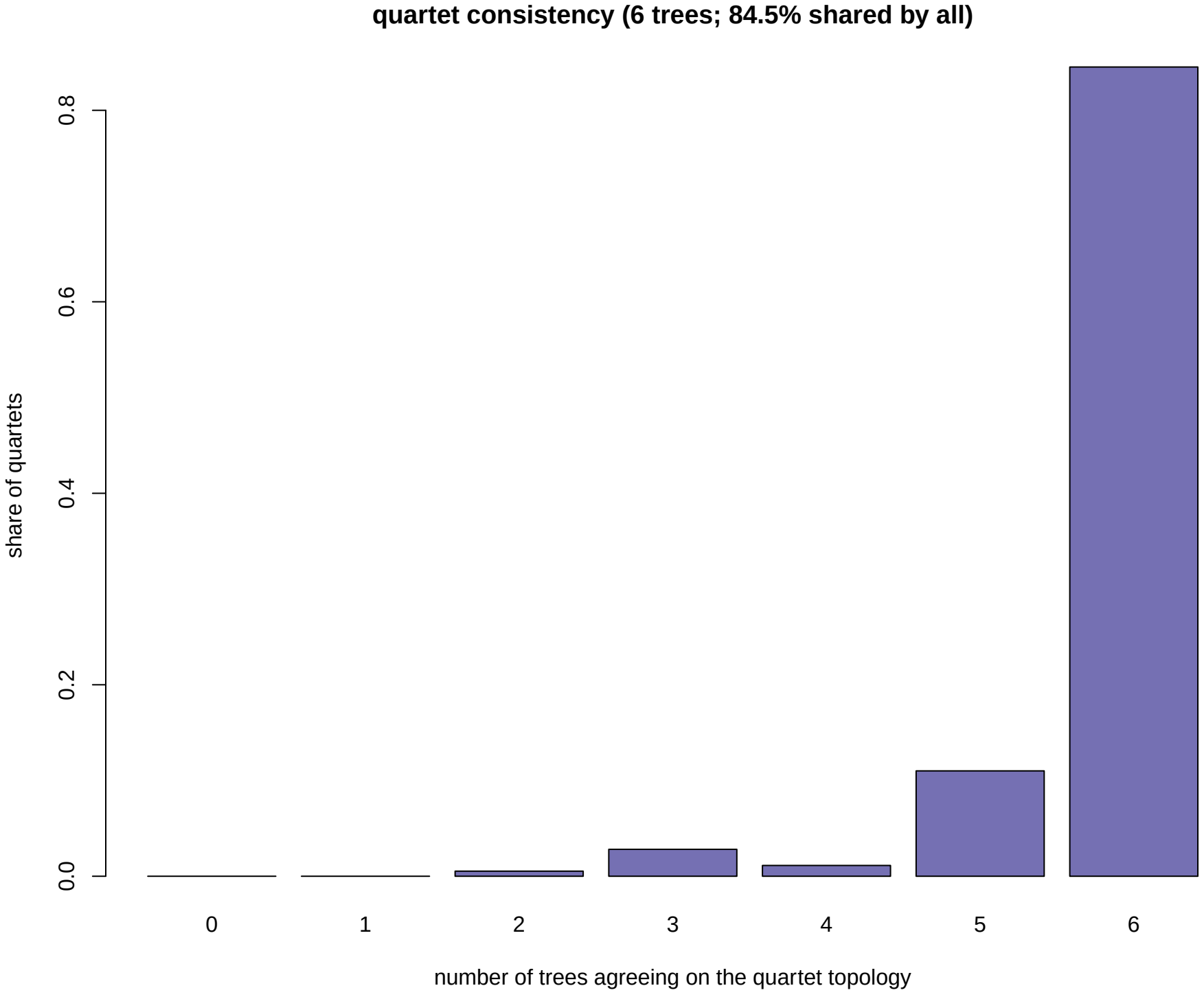
IQ-TREE vs AG-ncbi

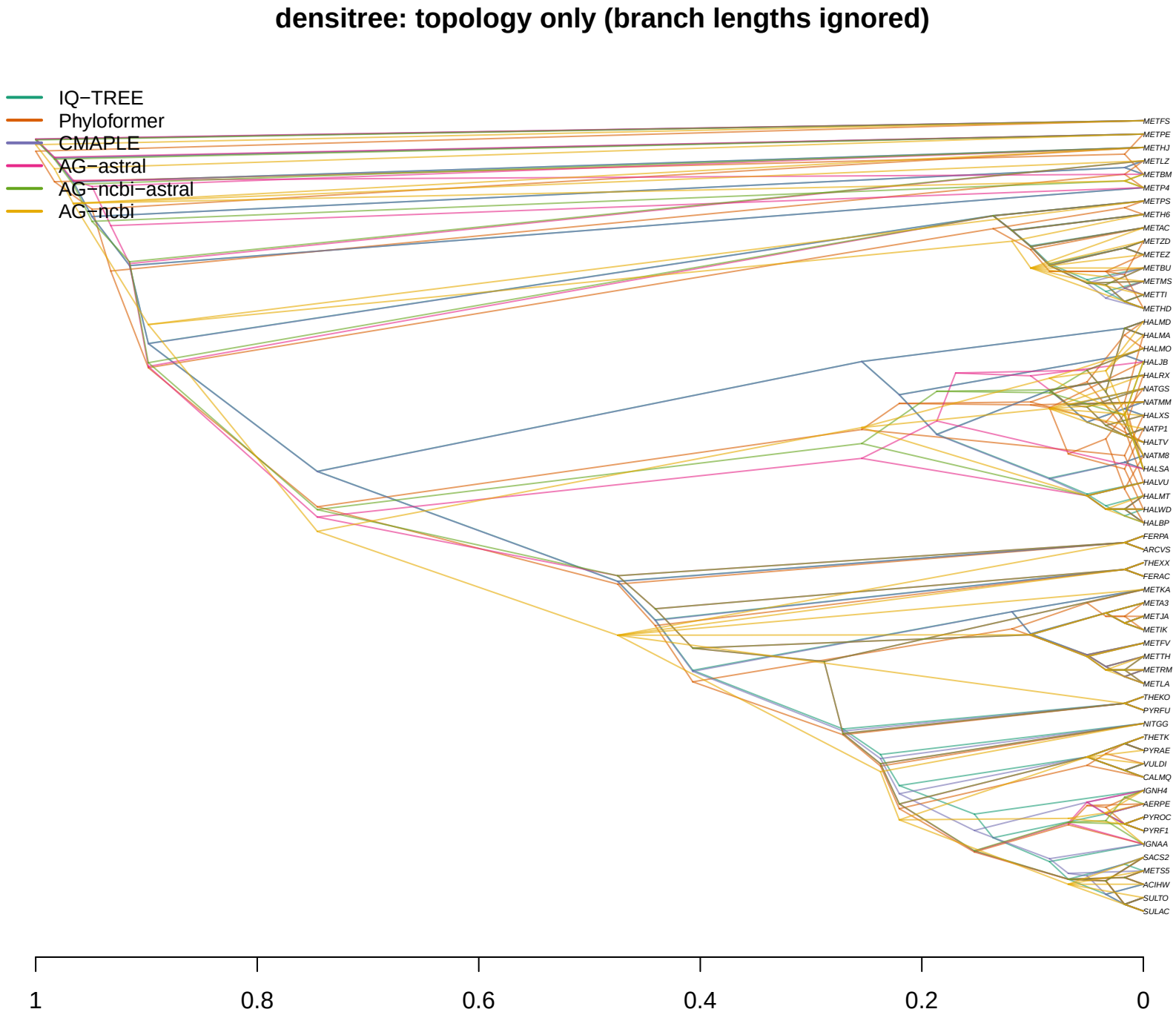


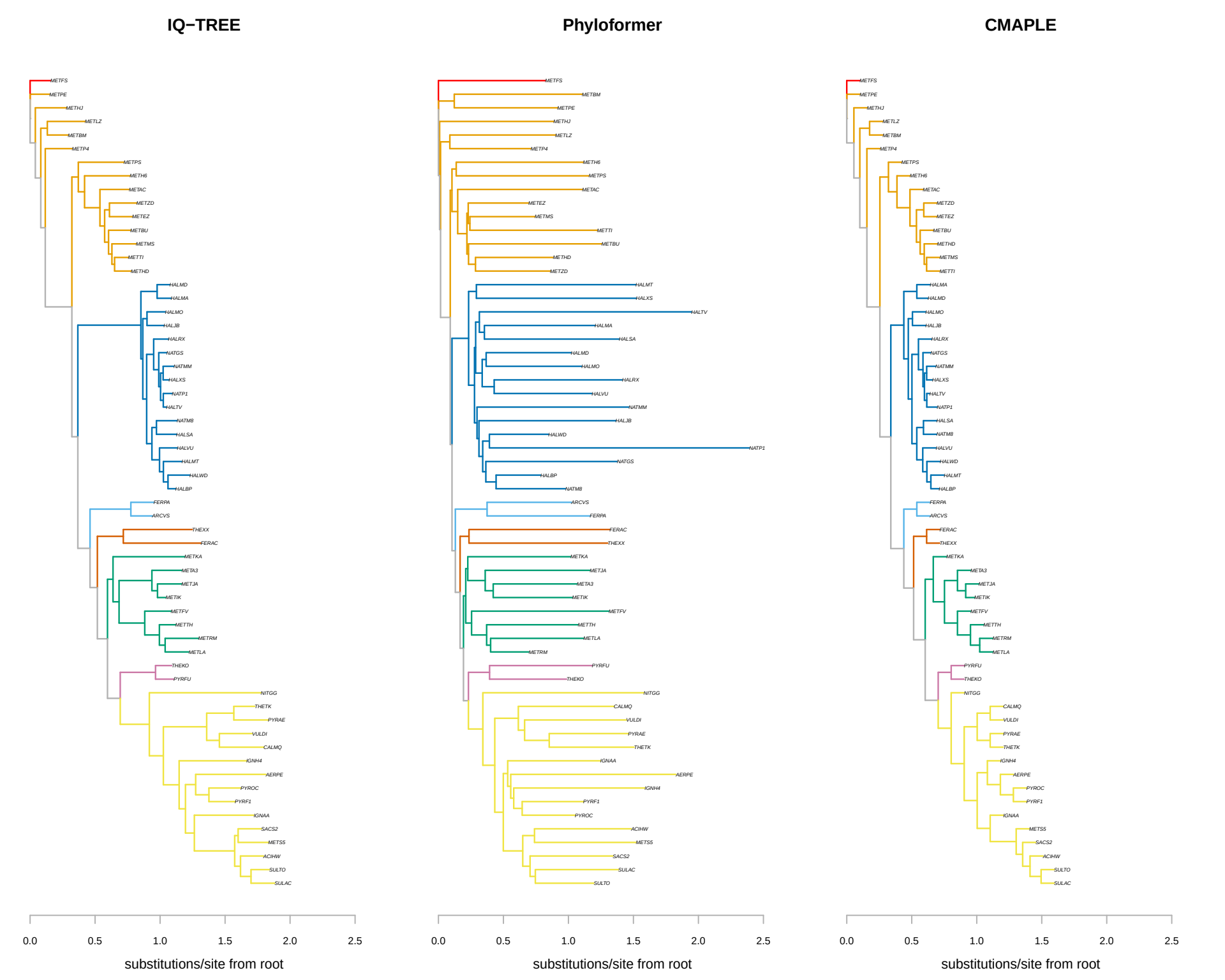

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
Phyloformer
AG−ncbi
AG−ncbi−astral
AG−astral
CMAPLE
IQ−TREE
Phyloformer
AG−ncbi
AG−ncbi−astral
AG−astral
CMAPLE
IQ−TREE


methods by RF distance
AG−ncbi
Phyloformer
AG−ncbi−astral
AG−astral
CMAPLE
IQ−TREE

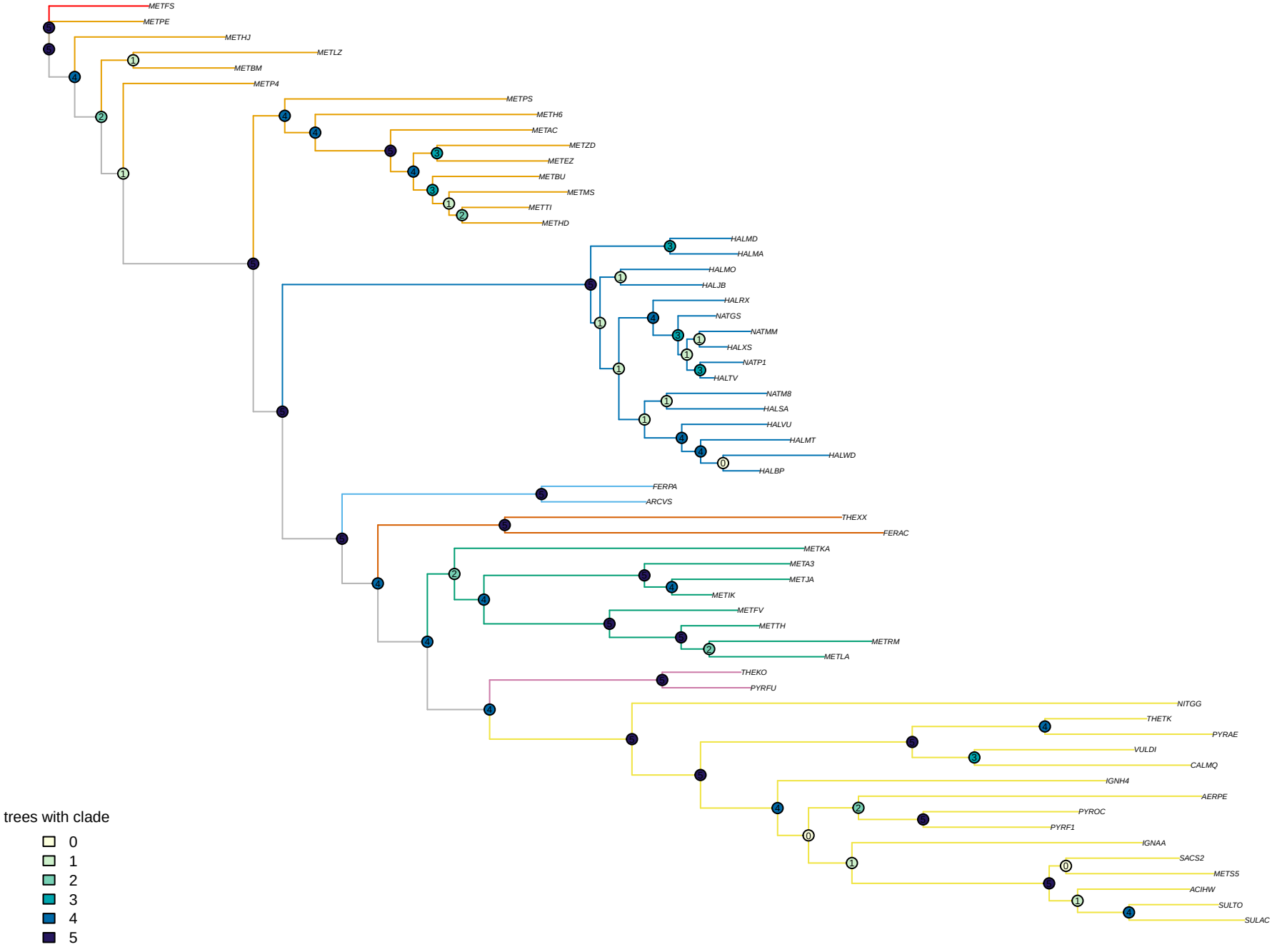

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
Phyloformer
CMAPLE
IQ−TREE
AG−ncbi
AG−ncbi−astral
AG−astral
Phyloformer
CMAPLE
IQ−TREE
AG−ncbi
AG−ncbi−astral
AG−astral


quartet consistency (6 trees; 84.5% shared by all)
share of quartets
0.0
0.2
0.4
0.6
0.8
0
1
2
3
4
5
6
number of trees agreeing on the quartet topology

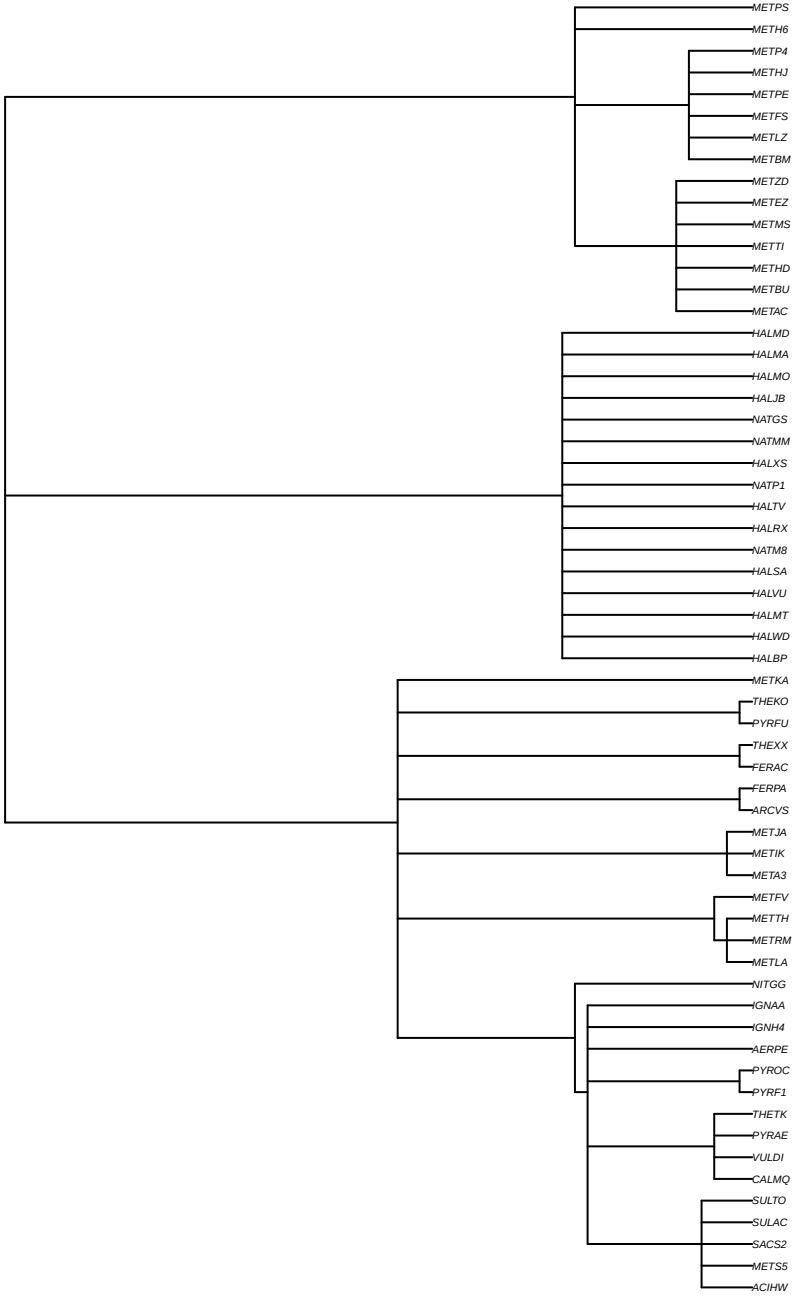

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
AG−astral
AG−ncbi−astral
AG−ncbi
METFS
METPE
METHJ
METLZ
METBM
METP4
METPS
METH6
METAC
METZD
METEZ
METBU
METMS
METTI
METHD
HALMD
HALMA
HALMO
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HALRX
NATGS
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PYRFU
NITGG
THETK
PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
SACS2
METS5
ACIHW
SULTO
SULAC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
METFS
METPE
METHJ
METLZ
METBM
METP4
METPS
METH6
METAC
METZD
METEZ
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PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
SACS2
METS5
ACIHW
SULTO
SULAC
0.0
0.5
1.0
1.5
2.0
2.5
substitutions/site from root

IQ−TREE clades: number of the other 5 trees that contain each clade



strict



majority rule

